Preliminary report on the emotion classifier project

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Abstract

In this report we share our approach to the facial emotion recognition problem. The main focus of this paper is to give the reader a brief overview over the datasets, the data preprocessing and the model architecture used in our approach.

1. Introduction

[TODO: Write Introduction]

2. Datasets

[TODO: Write Datasets section]

3. Methodology: Data Preprocessing

3.1. Approach and Quality Control

We prioritize a large dataset, but try to maintain quality. We clean unconstrained images to remove noise that hurts model training.

3.2. Detection and Geometric Filtering

We use MediaPipe for face detection, increasing the size of low-resolution images for accuracy, because MediaPipe was trained on bigger images.
Occlusion: Images are removed if no face is found or if key face parts (both eyes, nose) are blocked.
Pose Constraints: We reject images with a Yaw (left/right) rotation over 45 degrees. Pitch (up/down) and Roll (tilt) are also limited to avoid extreme angles, ensuring features are visible.

3.3. Backgrounds and Image Quality

Cropping: If the face is small ($< 10\%$ of the image), we crop to center it. We keep some background to help the model handle environmental noise.
Blur & Deduplication: We remove blurry images using the **Laplacian method** [1], this will also remove cartoon images and we remove exact duplicates using **SHA-256 hashing**, or **Perceptual Hashing** if more performance is needed.

3.4. Imbalance and Reduction of Background Noise

To reduce the effect of background noise, we will experiment using an **Attention-Hybrid CNN**. This architecture learns to focus on facial features over background noise. We handle the imbalance in emotion classes using **Class-Balanced Focal Loss** [2] instead of reducing the data, further more about it.

4. Model Architectures

To address the facial emotion recognition task (FER), our group investigates multiple convolutional neural networks (CNNs) that have demonstrated strong performance in image classification and facial analysis tasks. CNNs build the base for most of our models. This decision is based on according literature emphasizing the clear performance advantage of CNNs compared to other approaches such as traditional machine learning models [3]. Each team member focuses on implementing and evaluating selected architectures in order to identify the most suitable model for achieving a high classification accuracy while maintaining robust generalization performance.

- Dmytro Dumanskyy:
- Tiago Kienzle Hagen: ResNet, CERN, Poster++
- Aleksandar Neshov: ResNet, EfficientNet, DenseNet
- Daniel Pfefferkorn: GoogLeNet, VGG-19, EmoNeXt

Maximizing the accuracy of the FER task represents the core goal of our group. Apart from the aforementioned aspects, the actual architecture of our neural network plays a key role in achieving this goal to the best of our ability. In the follwoing section we want to highlight our decision making process.

4.1. Criteria for model selection

The suitability of a model for this project is evaluated based on the following criteria:

Firstly the design should be well documented in the literature to ensure scientifically proven effectiveness. For a model to be taken into closer consideration it should have reported strong results on standart benchmarks.

Secondly the architecture should be capable of extractig important facial featrures such as the eyes, the eyebrows and the mouth meaningfully.

Finally models that are prone to overfitting are seen as unfavorable for our project. The chosen network should be able to learn distinctive features well while not overly relying on a high number of samples to achieve acceptable results. Since FER datasets are limited in sample size, we will prioritize models that shine through their data efficiency.

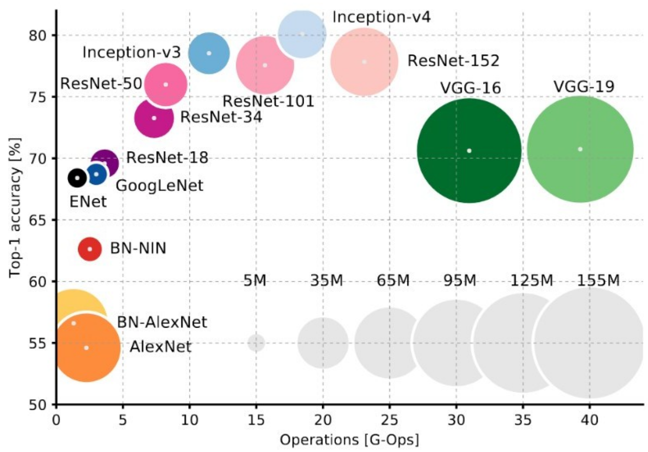

Figure 1. This figure from Kopalidis et al. [4] was used as a general reference as to which models to take into consideration.

4.2. Excluded model architectures

Not all types of models were categorized as suitable for the task at hand. Vision Transformers (ViT) are highly data hungry and tyically rely on large-scale pretraining, making them suboptimal for training from scratch on limited FER datasets. Furthermore any outdated neural networks that are no longer competitive in terms of accuracy or efficiency were used as baselines for comparison.

4.3. Training setup

We decided to research common training parameters used in scientific papers that yielded good results. All models are trained from scratch using a consistent training protocol to ensure fair comparison.

We decided to use Cross-Entropy loss as our loss function since it's widely used in similar classification tasks. Using Stochastic gradient descent (SGD) as our optimizer we are going to train the networks for approximately 30-50 epochs depending on individual model convergence. Furthermore we plan on exploring attention mechanisms to improve robustness against facial occlusions.

To ensure a consistent baseline fo rmodel evaluation all models are trained and tested on a predefined dataset.

4.4. Model selection strategy

We are currently in theprocess of deciding on a final neural network design for the entire group. Each team member is implementing their own model based on the architectures described above. The final model will be decided through the highest test accuracy on our collective test set.

5. Explainable AI

In order to portray the model's regions of interest in the image we plan on using GradCAM, which is a common method used for visualizing and interpreting networks by highlighting the image regions that contribute most strongly to a model's prediction.

References

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